measures of despersion

① Range ② variance ③ Standard deviation

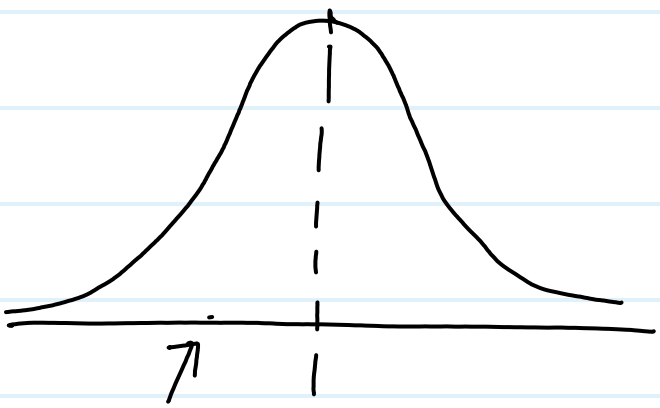
① Range =

$$[4, 3, 2, 0, 1, 5, 7, 6, 8, 10, 12, 13]$$

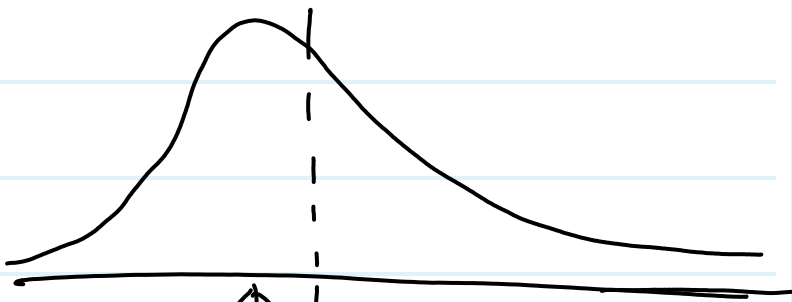
$x_{max} = 13$ $x_{min} = 0$

range = $13 - 0 = 13$

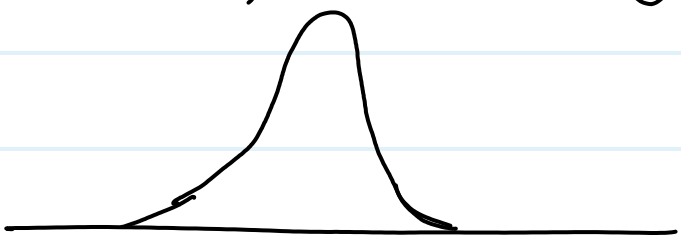
② variance ⇒



standard variablity



high variablity



low variablity

pop = N mean = μ

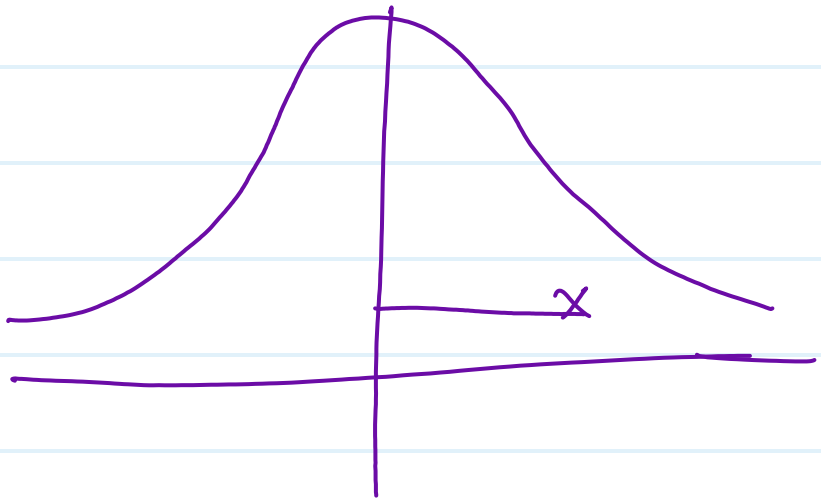
sample = n mean = $\bar{x}$

(pop) $\sigma^2 = \dfrac{\sum_{i=1} (x_i - \mu)^2}{n}$ ✓

(Sample) $s^2 = \dfrac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$ ✓

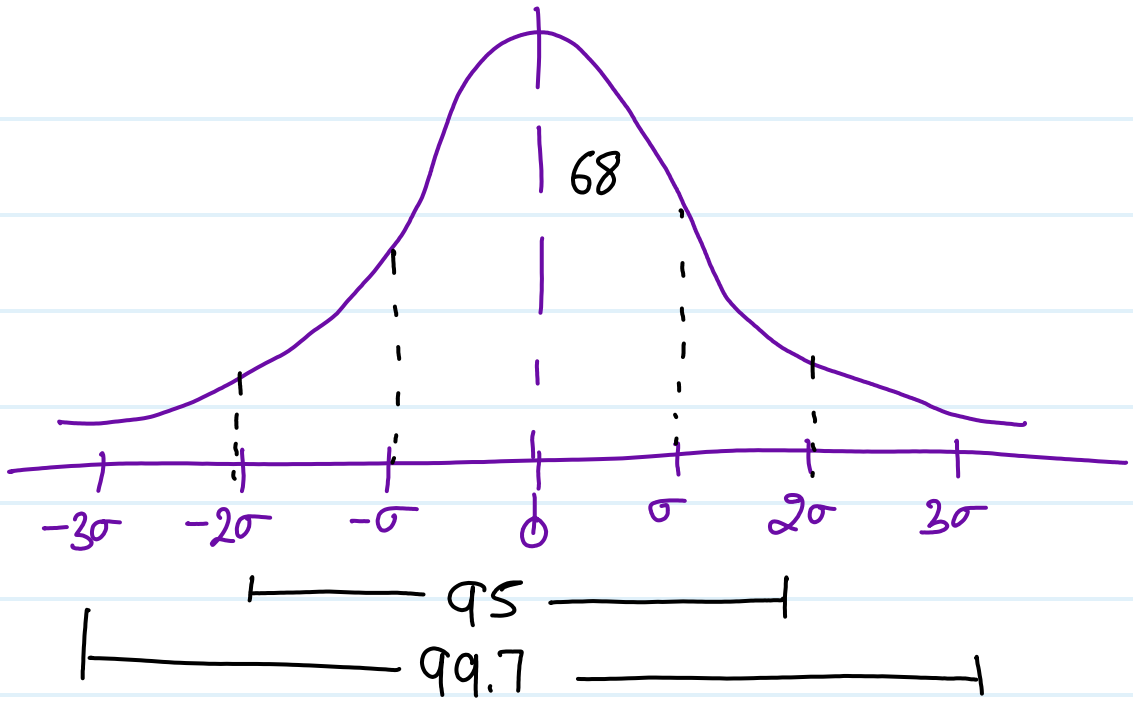
$(n-1)$ is degree of freedom

③ standard deviation →



pop $\sigma = \sqrt{\dfrac{\sum (x_i - \mu)^2}{n}}$

sample $s = \sqrt{\dfrac{\sum (x_i - \bar{x})^2}{n-1}}$



Emperical rule of formula

[68- 95 - 99.7]

Dataset = [2, 3, 4, 6, 3, 2, 1, 7, 3, 8, 5]

$n = 11$

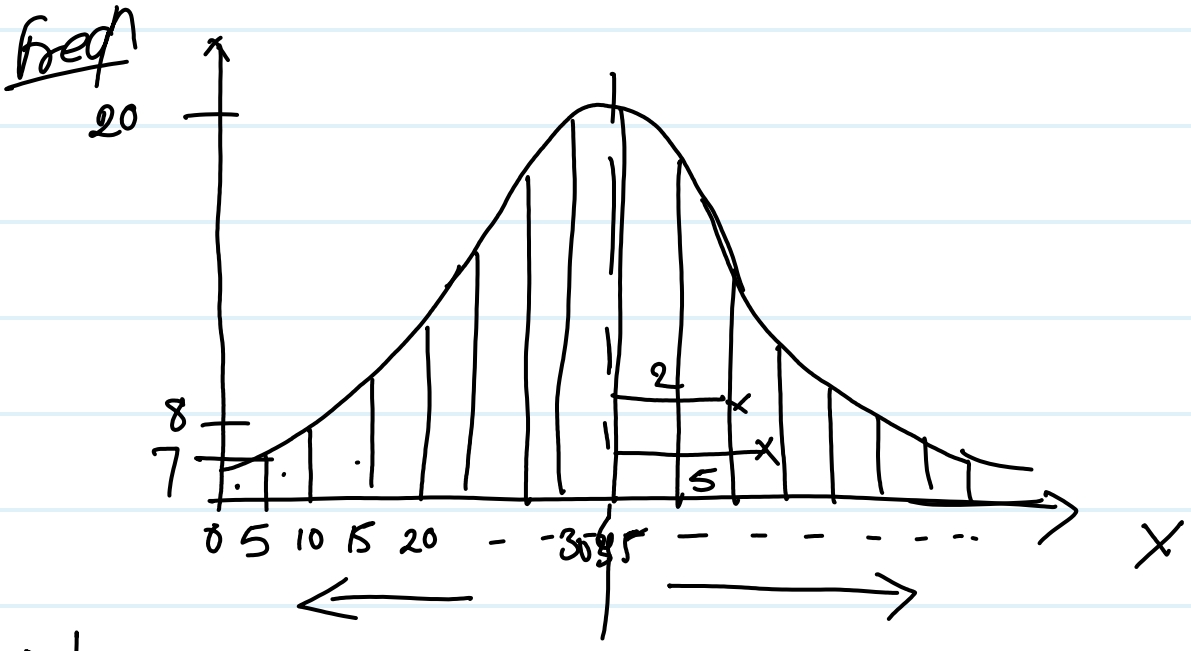
$\bar{x} = \frac{44}{11} = 4$

x_i	$\bar{x}$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$\sum (x_i - \bar{x})^2$
2	1	-2	4	
3		-1	1	14
4	4	0	0	
6		2	4	
3		-1	1	
2		-2	4	

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$= \sqrt{\frac{14}{11-1}} = \sqrt{\frac{14}{10}}$$

$$\sigma = \sqrt{1.4} \quad =)$$



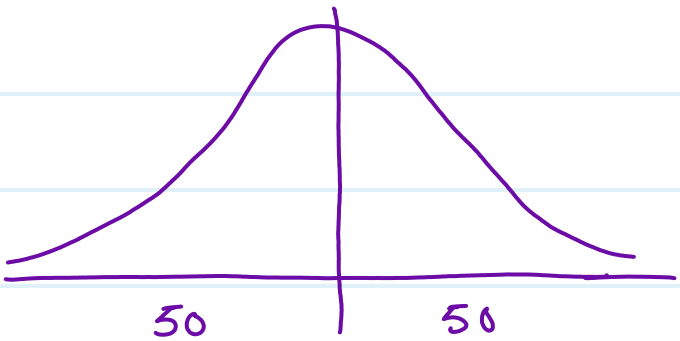
histogram

mean $= 4$ ✓

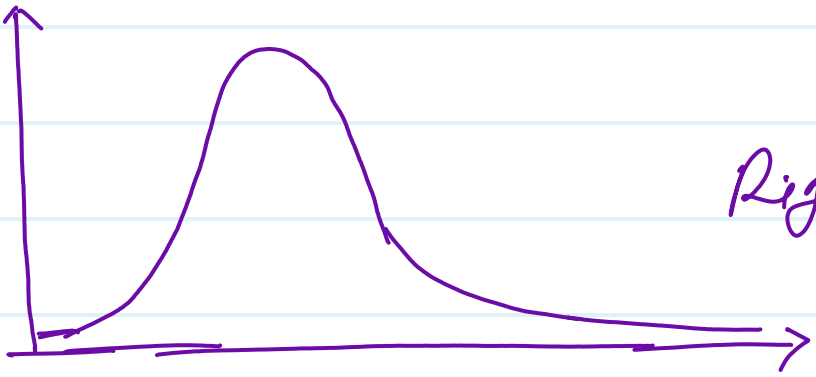
measures of shape

① skewness

zero

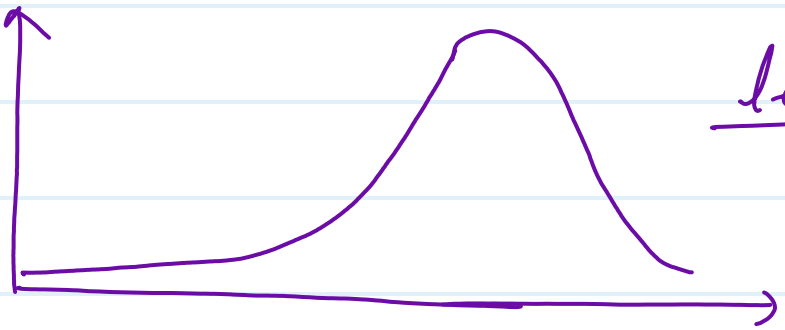


semmitric



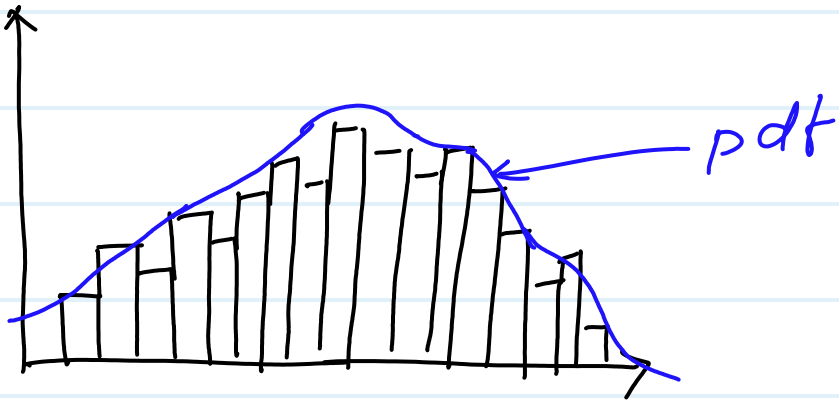
Right skewed

asemmitric



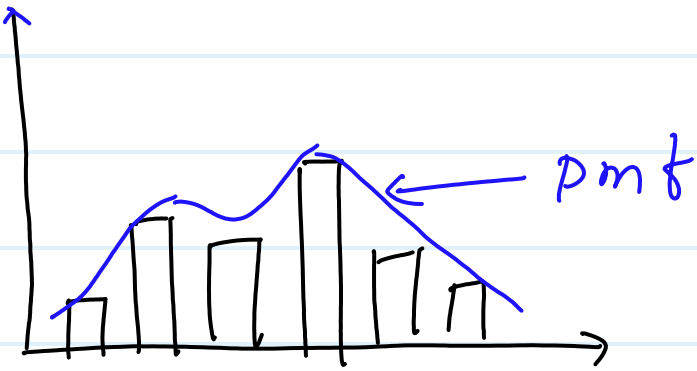
left skewed

a semmitric



Continous histogram →

✓ pdf = probability density func.



Descret histogram

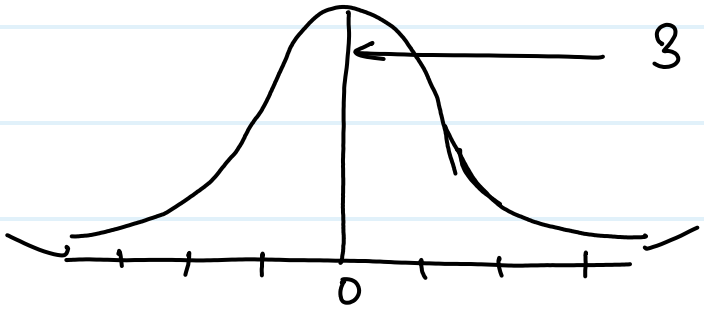
✓ pmf = probability mass func.

kde = kernel density esimator

② kurtosis (K)

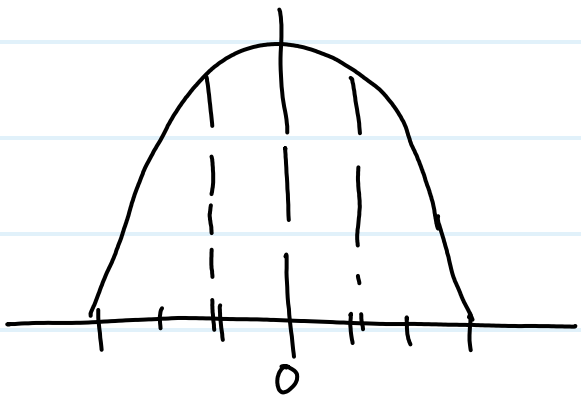
① mesokurtosis

$K = 3$



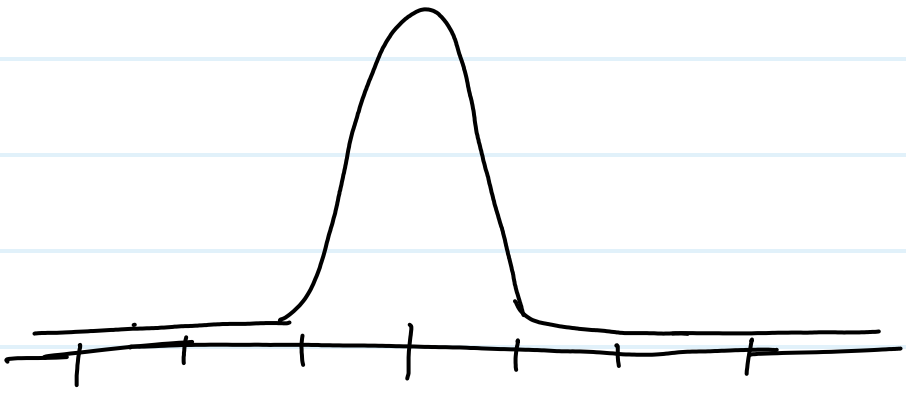
② platikurtosis

$K < 3$



③ Lepto kurtosis

$K > 3$



[4, 2, 5, 3, 6, 7, 16]

$\Rightarrow \frac{31}{7} = 4.3$ ✓

$\Rightarrow \frac{47}{7} = 6.5$ ✓

[4, 2, 5, _, 3, 6, 7, _, 16]

6.5 6.5 ✓

4.3 4.3

[2, 3, 4, 5, 6, 7, 16]

[10, 20, 30, 40, 100]

98

